Ophelia Chapman

opheliachapman1@gmail.com | (949)-299-7099 | Seattle, WA | U.S. Citizen

SUMMARY OF SKILLS:

▪ 3+ years working on mechanical design, product development, control systems, energy conversion, and project management across robotics, marine renewable energy, and data centers.
▪ SolidWorks certified; proficient in CAD and simulation (SolidWorks, Onshape, Ansys), controls platforms (Tridium Niagara, Direct Digital Control VisualLogic), and manufacturing processes (3D printing, machining, welding); experienced in GD&T and in creating controls and product-specific drawings; strong hand calculations for structural and fluid analysis
▪ Strong communication, leadership, and organizational skills shown through focused and general project management and coordinating interdisciplinary teams

EDUCATION:

▪ University of Washington – B.S. in Mechanical Engineering (ABET) | 4th-Year, June 2027 Graduation
▪ GPA: 3.26 ▪ FE Exam scheduled for December 2026
▪ Relevant coursework: Heat Transfer, Thermodynamics, Fluid dynamics, Kinematics and Linkages Design, Mechanics of Thin Films, Control Systems/Frequency Analysis, Machine Design Analysis, Electrical Circuit Analysis, Solidworks course

EXPERIENCE:

HVAC Controls Engineer Intern - ATS Critical Facilities ***June 2026 – September 2026***

▪ Supported controls engineering and BMS integration (BACnet, Tridium Niagara) for HVAC systems on an active data center construction project. Learned HVAC controls mechanical design and software unit manager integration
▪ Redlined and retrofitted DDC panels and controls drawings, and developed control and floor plan design drawings for fan coil and water source heat pump units using client documents and sequence of operations
▪ Assisted with commissioning and troubleshooting, trained new hires on controls configuration, BACnet integration, and engineering workflows. Learned data center fluid controls, flow/head analysis for fluid-side of data hall

President - Washington Wave ***June 2026 – Present***

▪ Project manager and main point of contact between club members, subteam leads, and team advisors
▪ Monitors progress on club goals, working with technical lead and subteam for overall project scheduling
▪ Runs weekly general and leadership meetings, collaborates with subteam leads to guide each team
▪ Onboards new members into club, communicates with prospective members, manages application processes, integrates new members into subteams. Interim environmental subteam lead, working on ecosystem and WEC life-cycle analysis

Mechanical Team VP, Actuator Team Lead - Washington Wave ***Oct 2025 – June 2026***

▪ Led the design and integration of power take-off system converting mechanical wave motion into renewable electrical energy. Developed design, build, and test prototype and worked with electronics team for project integration and controls
▪ Led development of custom in-house designed and built linear actuator with stepper motor and manufactured housing. Completed software and hand-calculated fluid analysis on hydraulic pump to find required actuator force.
▪ Helped oversee subteam operations, managed project timelines, and onboarded new team members

Structural Design Engineer - Husky Robotics ***Oct 2023 – June 2025***

▪ Designed and modeled the rocker-bogie suspension and chassis for a large-scale Mars rover
▪ Assisted in design of 70+ chassis components and 50+ GD&T-dimensioned drawings, using SolidWorks FEA to validate structural performance, assisted manufacturing on bandsaw, CNC lathe, and welding
▪ Designed, 3D-printed, cast, and tested new wheel concepts for the 2026 Husky Robotics rover

ADDITIONAL ROLES:

Electrospray Propulsion R&D Engineer – Husky Satellite Lab ***September 2025 - June 2026***
Solid Mechanics R&D Engineer – iMatter Lab ***January 2025 - September 2025***